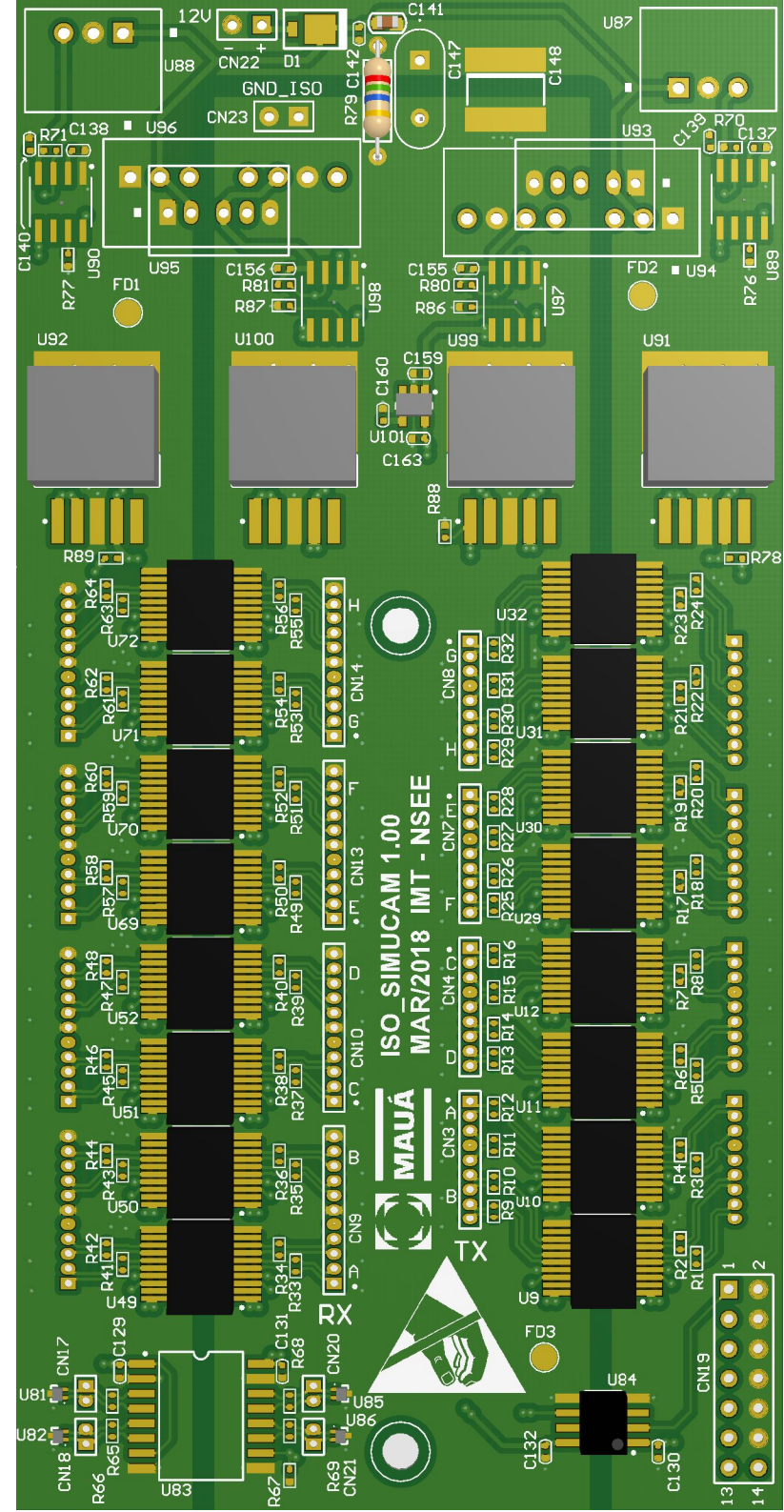




BOARD TITLE: ISO SIMUCAM

AUTHOR: CCB

CODE NUMBER:

REVISION: 1_00

LAYER STACKUP:

TOP / BOTTOM: GND PLANE + SIGNALS

REVISIONS DESCRIPTION:

1_00 - Initial revision - Mar - 2018

GENERAL NOTES:

1. LAMINATE SPECIFICATIONS:

- TOTAL THICKNESS: 1.0 mm +/- 0.1 mm
- LAYER NUMBERS: 2
- Cu THICKNESS (FINISHED): 1 oz / 1 oz
- RAW MATERIAL: FR-4
- TG MIN.: 150 °C
- TD MIN.: 310 °C

2. HOLE CHARACTERISTICS:

- PLATED HOLE TOLERANCE: +/- 0.1 mm
- NON PLATED HOLE TOLERANCE: +/- 0.1 mm
- DISPLACEMENT: 0.1 mm MAX.
- Cu THICKNESS ON HOLE: 20 um + SnPb MIN.

3. FINISHING:

- ## - HOT AIR SOLDER LEVELING (HASL) - SnPb

4. SILKSCREEN MASK:

- WHITE (THERMAL CURE EPOXY)

5. SOLDER MASK:

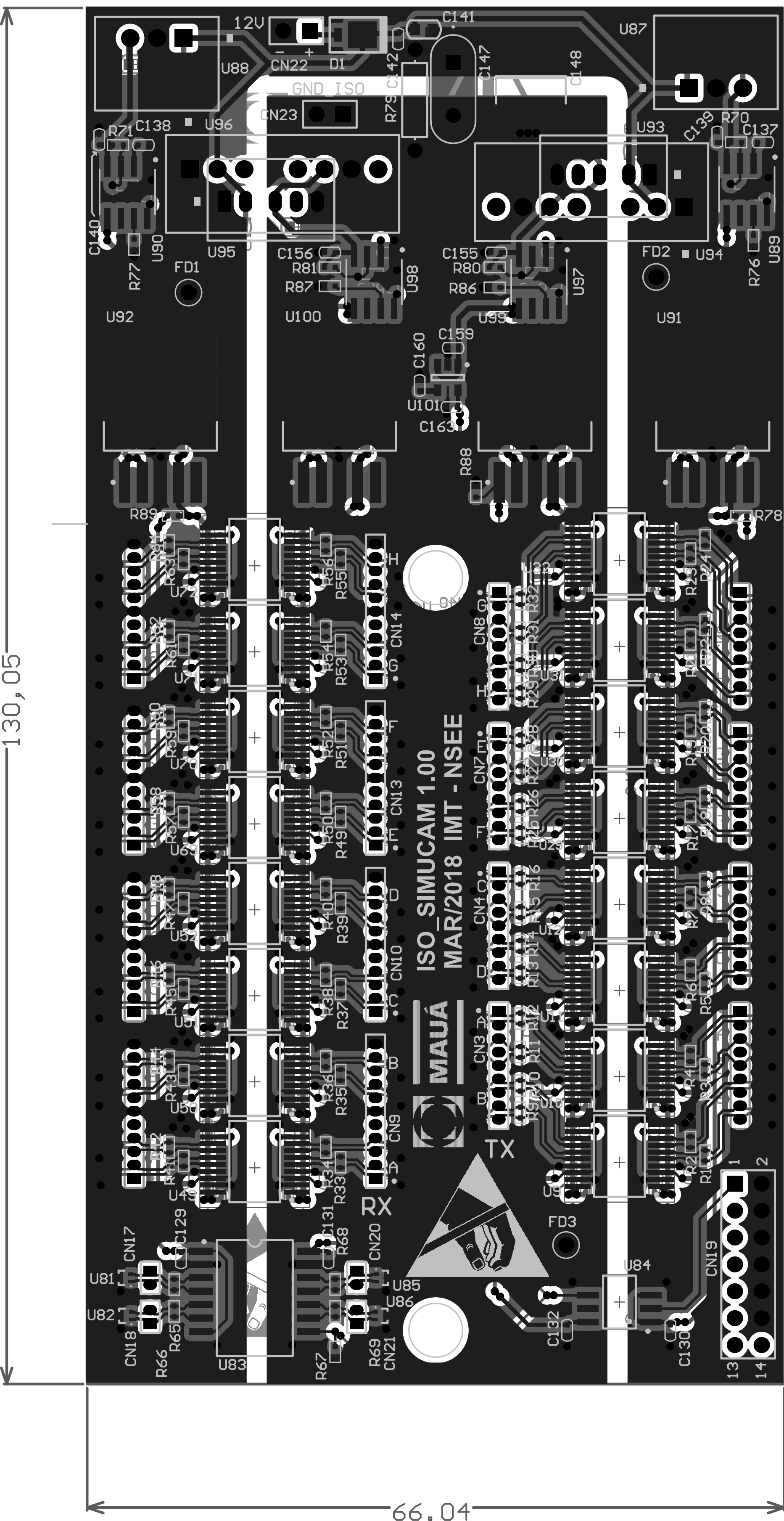
- BOTH SIDES
- GREEN EPOXY THERMAL CURE OR PHOTO IMAGEABLE
- IONIC CONTAMINATION AFTER SOLDERING: 14 ug NaCl / in2 MAX.
- COVER VIAS WITH SOLDER MASK ON BOTH SIDES
EXCEPT ON CN21 BORDERS, AS INDICATED IN GERBER

6. SPECIFIED STANDARDS:

- IPC-6011 CLASS 2 / IPC-6012 CLASS 2 / IPC-A-600 CLASS 2**

7. ADDITIONAL NOTES:

- DIMENSIONAL TOLERANCE: +/- 0.2 mm
- ELECTRICAL TESTS ARE MANDATORY
- GERBER MODIFICATIONS ARE ONLY ALLOWED
WITH EXPLICIT AUTHOR / RESPONSIBLE AUTHORIZATION



BOARD TITLE: ISO_SIMUCAM

AUTHOR: CCB

CODE NUMBER:

REVISION: 1_00

LAYER STACKUP:

TOP / BOTTOM: GND PLANE + SIGNALS

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REVISIONS DESCRIPTION:

1_00 - Initial revision - Mar - 2018

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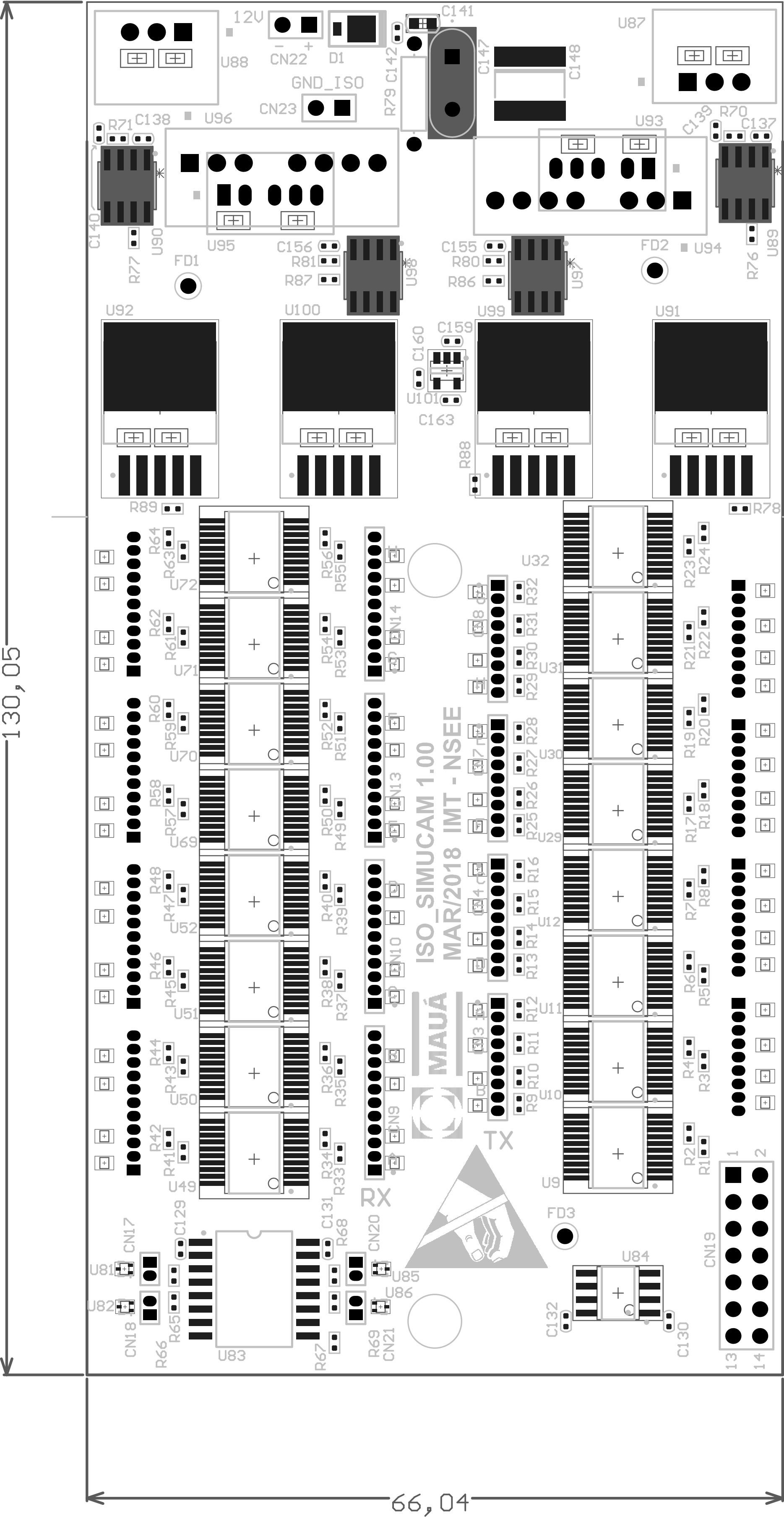
- BOTH SIDES
- GREEN EPOXY THERMAL CURE OR PHOTO IMAGEABLE
- IONIC CONTAMINATION AFTER SOLDERING: 14 ug NaCl / in2 MAX.
- COVER VIAS WITH SOLDER MASK ON BOTH SIDES EXCEPT ON CN21 BORDERS, AS INDICATED IN GERBER

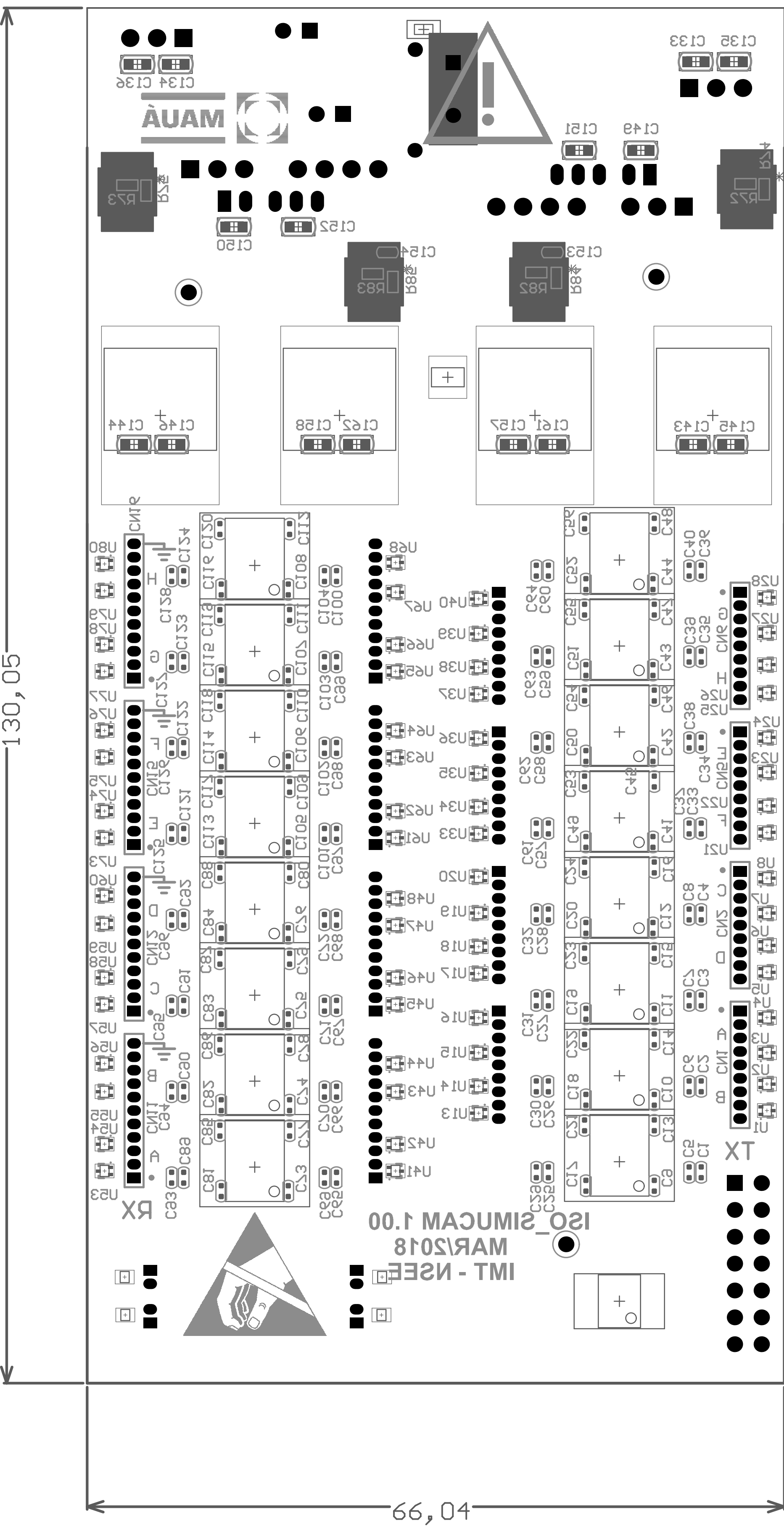
6. SPECIFIED STANDARDS:

- IPC-6011 CLASS 2 / IPC-6012 CLASS 2 / IPC-A-600 CLASS 2

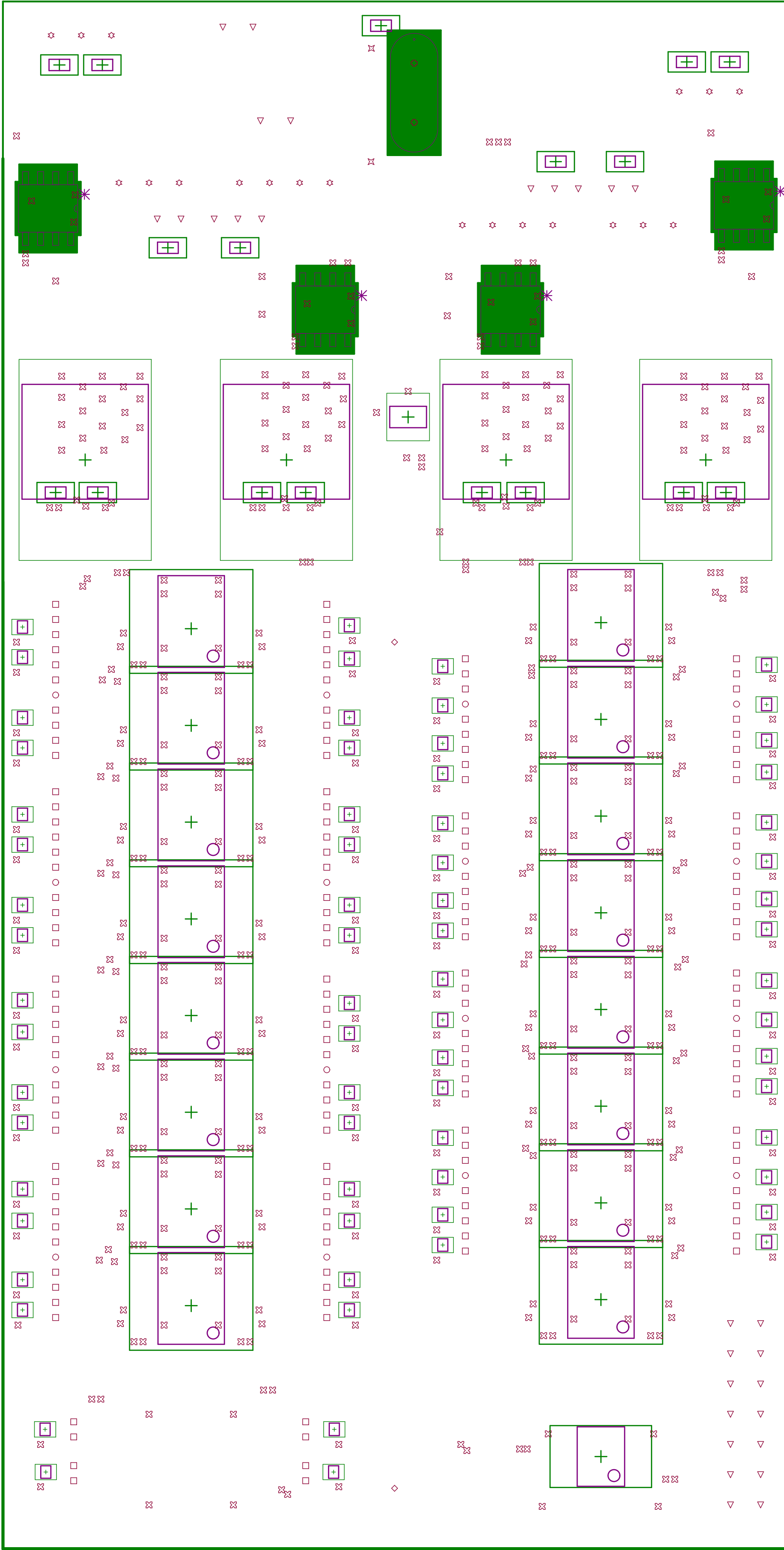
7. ADDITIONAL NOTES:

- DIMENSIONAL TOLERANCE: +/- 0.2 mm
- ELECTRICAL TESTS ARE MANDATORY
- GERBER MODIFICATIONS ARE ONLY ALLOWED WITH EXPLICIT AUTHOR / RESPONSIBLE AUTHORIZATION



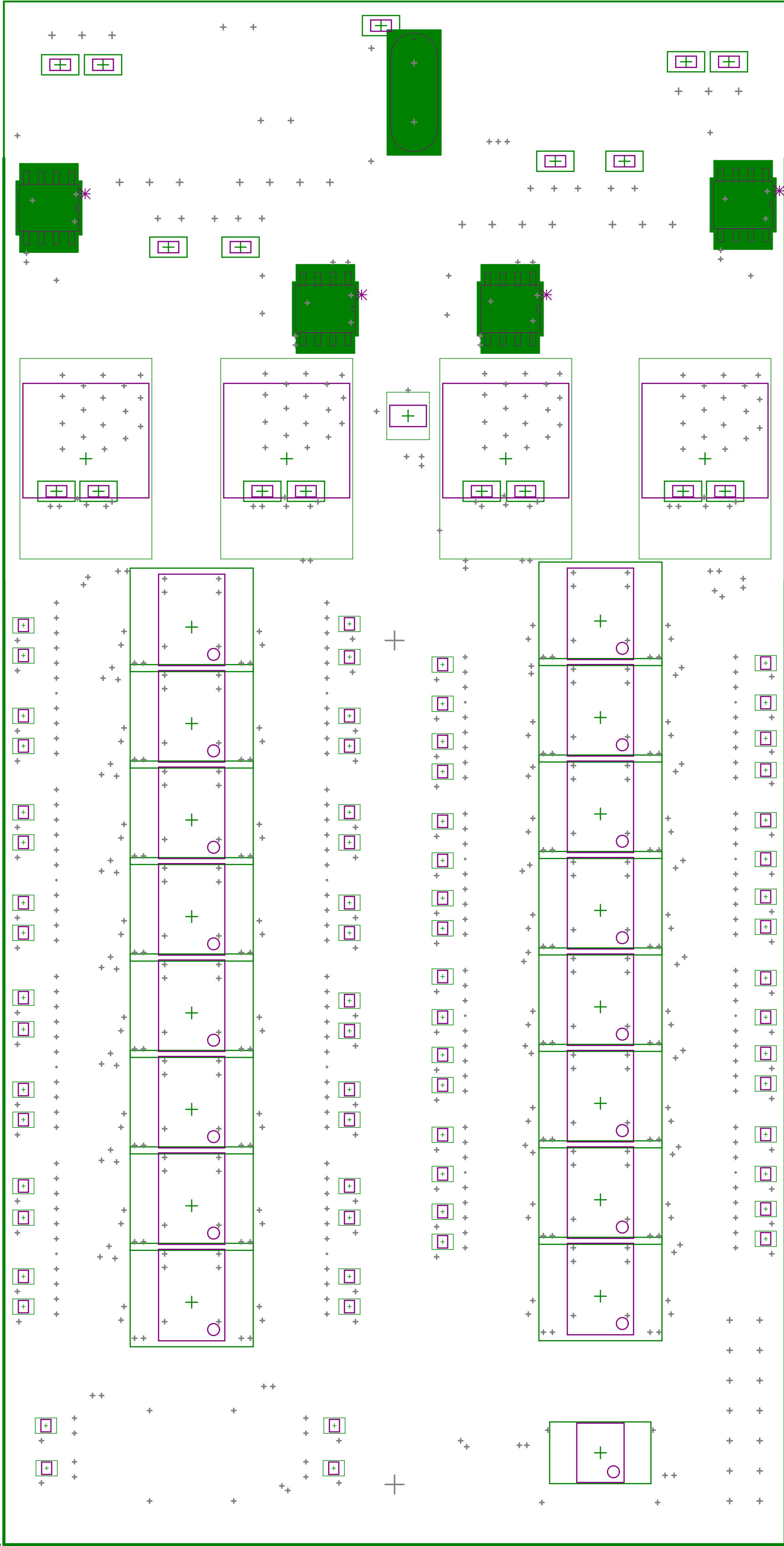


130,05



66,04

130,05



66,04

Comment	Description	Designator	Footprint	LibRef	Quantity	Value
errd	ATENCAO	A11	JSA_DOCUMENTACAO	ATENCAO	1	Atention
Capacitor Cerâmico SMD	Capacitor Cerâmico SMD	C1, C2, C3, C4, C9, C10, C11, C12, C17, C18, C19, C20, C29, C30, C31, C32, C33, C34, C36, C38, C41, C42, C43, C44, C49, C50, C51, C52, C61, C62, C63, C64, C66, C66, C67, C68, C73, C74, C75, C76, C81, C82, C83, C84, C93, C94, C96, C96, C97, C98, C99, C100, C105, C106, C107, C108, C113, C114, C115, C116, C125, C136, C137, C138	M402-cap	Capacitor Cerâmico SMD	64	10u
Capacitor Cerâmico SMD	Capacitor Cerâmico SMD	C5, C6, C7, C8, C13, C14, C15, C16, C21, C22, C23, C24, C25, C26, C27, C28, C37, C38, C39, C40, C45, C46, C47, C48, C53, C54, C55, C56, C57, C58, C59, C60, C69, C70, C71, C72, C77, C78, C79, C80, C85, C86, C87, C88, C89, C90, C91, C92, C101, C102, C103, C104, C109, C110, C111, C112, C117, C118, C119, C120, C121, C122, C123, C124, C129, C130, C131, C132, C139, C140, C142, C155, C156	M402-cap	Capacitor Cerâmico SMD	73	100n
Capacitor Cerâmico SMD	Capacitor Cerâmico SMD	C133, C134, C135, C136, C143, C143, C144, C145, C146, C149, C150, C151, C152, C157, C158, C165, C167	M403 - CAP - montagem na maquina	Capacitor Cerâmico SMD	17	10u
Capacitor Cerâmico SMD	Capacitor Cerâmico SMD	C137, C138, C153, C164	M402-cap	Capacitor Cerâmico SMD	4	1u
CHP330102A3B	Imported	C147	CAP_10K4	CHP330102A3B	1	10u @ 2KV
Capacitor Cerâmico SMD	Capacitor Cerâmico SMD	C148	C2225	Capacitor Cerâmico SMD	1	100n @ 2KV
Capacitor Cerâmico SMD	Capacitor Cerâmico SMD	C159, C163	M402-cap	Capacitor Cerâmico SMD	24	7u
Capacitor Cerâmico SMD	Capacitor Cerâmico SMD	C160	M402-cap	Capacitor Cerâmico SMD	1	470p
MHDR129	Header, 9-Pin	CN1, CN2, CN3, CN4, CN5, CN6, CN7, CN8	MHDR129 Axom	MHDR129	8	
MHDR121	Header, 12-Pin	CN9, CN10, CN11, CN12, CN13, CN14, CN15, CN16	MHDR121 Axom	MHDR121	8	
MHDR1X2	Header, 2-Pin	CN17, CN18, CN20, CN21	MHDR1X2 AxomN	MHDR1X2	4	Header 1.27 mm - 2 vias
Header 7X2	Header, 7-Pin, Dual row	CN19	HDR2X7	Header 7X2	1	
Header 2	Header, 2-Pin	CN22, CN23	MHDR1X2	Header 2	2	Header 2.54 mm
NPVBW130L735	Surface Mount Schottky Power Rectifier - 1.0 AMPERES, 30 VOLTS	D1	DO-216AA	NPVBW130L735	1	
errd	ESD	E20	ES7ATICA	ESD	1	
errd	ESD	E201	ESTATICA	ESD	1	ESD logo
Comment		R61	TEXTO FABRICACAO	Texto de fabricação	1	
errd	Reductal	R01, F02, F03	REDUCAL	Reductal	3	
Res3	Resistor	R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R68, R69	M402-res	Res3	68	100R
Res3	Resistor	R67, R68	M402-res	Res3	2	10k
Res3	Resistor	R70, R71, R76, R80, R81, R89	M402-res	Res3	1	50R
Res3	Resistor	R72, R73, R83, R83	M402-res	Res3	4	15k @ 1%
Res3	Resistor	R74, R75, R84, R85	M402-res	Res3	4	47k @ 1%
Res3	Resistor	R76, R77, R86, R87	M402-res	Res3	4	10k
Res3	Resistor	R79	CR25	Res3	1	1M @ PTH - 1/4W - 1.6 kV
TP02000RTR	ESD Protection Array for High-Speed Data Interfaces, 2 Channels, -40 to +85 degC, 3-pin SOT (DRT), Green (RoHS & no Sb/Br)	U1, U2, U3, U4, U5, U6, U7, U8, U13, U14, U15, U16, U17, U18, U19, U20, U21, U22, U23, U24, U25, U26, U27, U28, U33, U34, U35, U36, U37, U38, U39, U40, U41, U42, U43, U44, U45, U46, U47, U48, U53, U54, U55, U56, U57, U58, U59, U60, U61, U62, U63, U64, U65, U66, U67, U68, U73, U74, U75, U76, U77, U78, U79, U80, U81, U82, U85, U86	DR13	TP02000RTR	68	
ADN4800R12	3.3V BMM3.75-VV PWR: 480 Mbps, Dual-Channel LVDS Isolators	U9, U10, U11, U22, U28, U31, U32, U49, U50, U51, U52, U69, U70, U71, U72	SDP-RS-20	ADN4800R12	16	
ISOT7ELL_CM	High-Performance, 8000-VIN Reinforced Isolated Dual LVDS Buffer	U93	SDAC-18_CM	ISOT7ELL_CM	1	
ISOT71PD	High-Speed, Robust EMC Reinforced Single-Channel Digital Isolator	U94	SDIC-B	ISOT71PD	1	
R-78E5.0-1.0	1-Terminal Positive dc-dc regulator - 5V @ 1A	U97, U98	TD-230	R-78E5.0-1.0	2	5V - 1A
TPS25921AOR	Imported	U99, U99, U97, U98	D8	TPS25921AOR	4	
MCP1826-3302/ET	1000 mA, Low Voltage, Low Quiescent Current LDO Regulator, 5-Pin DOPAK, Extended Temperature	U91, U92, U99, U100	DDPAK-ET5_N	MCP1826-3302/ET	4	
TPN31211	1-Terminal Positive dc-dc iso regulator - 5V @ 0.6A	U93, U96	TPN3	TPN31211	2	5V - 0.6A
R56-12055	1-Terminal Positive dc-dc iso regulator - 5V @ 1.2A	U94, U96	R56	R56-12055	2	5V - 1.2A
TC1014-3.3VCT713	50 mA, 100 mA and 150 mA CMOS LDOs with Active-Low Shutdown and Reference Bypass, 5-Pin SOT-23, Extended Temperature, Tape and Reel	U101	SOT-23 CT5_N	TC1014-3.3VCT713	1	

Design Rules Verification Report

Filename : C:\projetos\simucam_iso\hw\pim_iso_simucam\pcb\iso_simucam.PcbDoc

Warnings 6
Rule Violations 0

Warnings	
Zero hole size multi-layer pad(s) detected	3
Multilayer Pads with 0 size Hole found	3
Total	6

Rule Violations	
Clearance Constraint (Gap=20mil) (InNamedPolygon('gnd_iso_bottom')),(All)	0
Clearance Constraint (Gap=20mil) (InNamedPolygon('gnd_iso_top')),(All)	0
Clearance Constraint (Gap=20mil) (InNamedPolygon('gnd_bottom')),(All)	0
Clearance Constraint (Gap=8mil) (All),(All)	0
Clearance Constraint (Gap=20mil) (InNamedPolygon('gnd_top')),(All)	0
Clearance Constraint (Gap=75mil) (InNamedPolygon('gnd_top')),(InNamedPolygon('gnd_iso_top'))	0
Clearance Constraint (Gap=60mil) (HoleDiameter = 125;),(InNamedPolygon('gnd_iso_bottom'))	0
Clearance Constraint (Gap=60mil) (HoleDiameter = 125;),(InNamedPolygon('gnd_iso_top'))	0
Clearance Constraint (Gap=7mil) (HasFootprint('SSOP-RS-20')),(All)	0
Clearance Constraint (Gap=75mil) (InNamedPolygon('gnd_bottom')),(InNamedPolygon('gnd_iso_bottom'))	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0
Un-Routed Net Constraint (All)	0
Width Constraint (Min=10mil) (Max=20mil) (Preferred=15mil) (All)	0
Power Plane Connect Rule(Direct Connect)(Expansion=20mil) (Conductor Width=10mil) (Air Gap=10mil) (Entries=4)	0
Hole Size Constraint (Min=0mil) (Max=500mil) (All)	0
Hole To Hole Clearance (Gap=10mil) (All),(All)	0
Minimum Solder Mask Sliver (Gap=0mil) (All),(All)	0
Silk To Solder Mask (Clearance=0mil) (IsPad),(All)	0
Silk to Silk (Clearance=2mil) (All),(All)	0
Net Antennae (Tolerance=0mil) (All)	0
Height Constraint (Min=0mil) (Max=1000mil) (Preferred=500mil) (All)	0
Total	0

Zero hole size multi-layer pad(s) detected	
Pad FD1-0(378mil,4061mil) on Multi-Layer on Net GND	
Pad FD2-0(2122mil,4119mil) on Multi-Layer on Net GND	
Pad FD3-0(1786mil,517mil) on Multi-Layer on Net GND_ISO	

Multilayer Pads with 0 size Hole found	
Pad FD1-0(378mil,4061mil) on Multi-Layer	
Pad FD2-0(2122mil,4119mil) on Multi-Layer	
Pad FD3-0(1786mil,517mil) on Multi-Layer	

Electrical Rules Check Report

Class	Document	Message
		Successful Compile for iso_simucam.PrjPCB